

GenAI. Lecture 1

A. Ryzhikov¹, S. Ali¹, A. Akhmatova¹, T. Ramazyan¹, K. Aithadzhaev¹, A. Khizhik¹

¹National Research University Higher School of Economics



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LAMBDA • HSE

GenAI problem



Generative AI

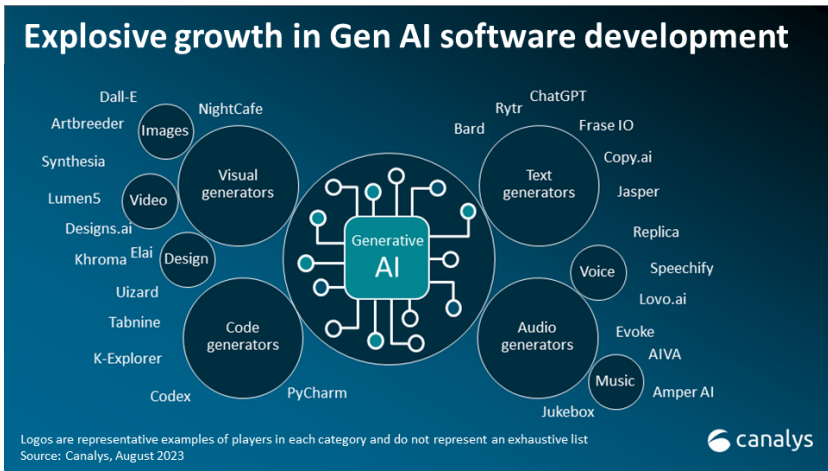


Figure: Generative AI applications

Problem Statement



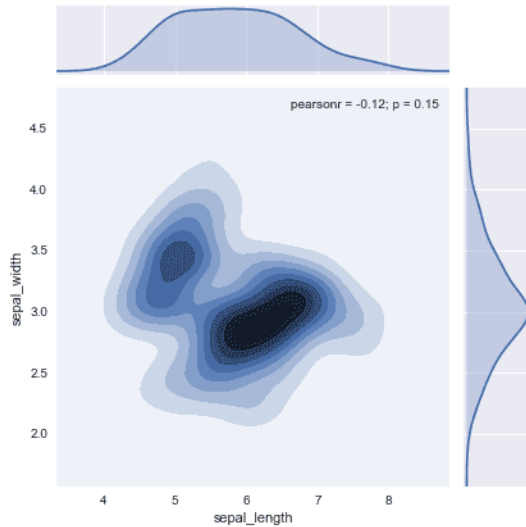
Formal statement

$P(X)$, $P(X|Y)$ - Generative AI (unconditional, conditional)

$P(Y|X)$ - Predictive AI

So, the problem is learn $P(X)$ (namely, to sample $X \sim P(X)$).

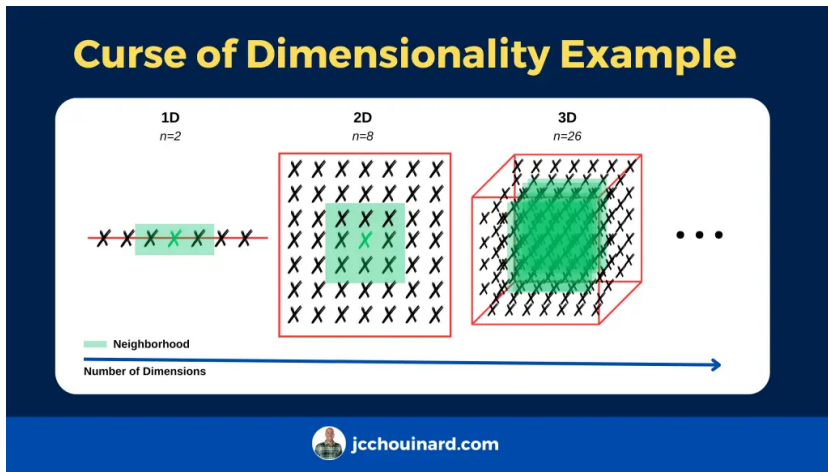
2D case



Main challenges

- ▶ Multimodality
- ▶ Dimensionality curse

Dimensionality curse



How to learn in higher dimensions?

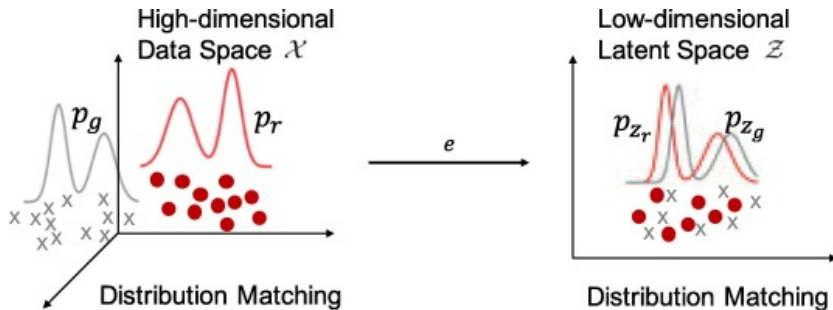
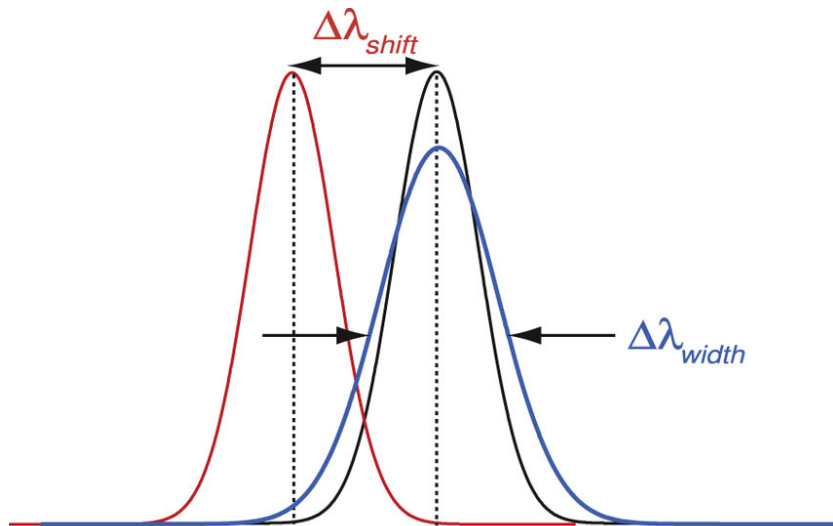


Figure: Low-dimensional (latent) distribution matching

Which objective to minimize?



Distances



Total Variation distance

For $p(x)$ and $q_\theta(x)$ being PDFs:

$$D(p(x), q_\theta(x)) = \frac{1}{2} \int |p(x) - q_\theta(x)| dx$$

This can be rewritten using Scheffé's theorem:

Scheffé's Theorem

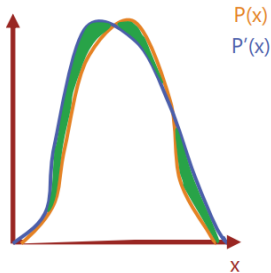
$$D(p(x), q_\theta(x)) = \sup_A \left| \int_A p(x) dx - \int_A q_\theta(x) dx \right|$$

Where A is any measurable set.

Total Variation distance

- ▶ symmetric
- ▶ interpretable

KL-divergence



$$\int P(x) \ln \frac{P(x)}{P'(x)} dx$$

JS-divergence

For $p(x)$ and $q(x)$, two probability distributions:

$$JS(p, q) = \frac{1}{2} \left(KL(p(x) \parallel \frac{p(x) + q_\theta(x)}{2}) + KL(q_\theta(x) \parallel \frac{p(x) + q_\theta(x)}{2}) \right)$$

- ▶ Symmetric
- ▶ Nonnegative:

$$0 \leq JS(P, Q) \leq \ln(2)$$

- ▶ Can be transformed to a true distance:

$$\sqrt{JS(p, q)}$$

f-divergence

$$D_f(p, q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx$$

Name	$D_f(P\ Q)$	Generator $f(u)$
Total variation	$\frac{1}{2} \int p(x) - q(x) dx$	$\frac{1}{2} u - 1 $
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} dx$	$u \log u$
Reverse Kullback-Leibler	$\int q(x) \log \frac{q(x)}{p(x)} dx$	$-\log u$
Pearson χ^2	$\int \frac{(q(x) - p(x))^2}{p(x)} dx$	$(u - 1)^2$
Neyman χ^2	$\int \frac{(p(x) - q(x))^2}{q(x)} dx$	$\frac{(1-u)^2}{u}$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 dx$	$(\sqrt{u} - 1)^2$
Jeffrey	$\int (p(x) - q(x)) \log \left(\frac{p(x)}{q(x)} \right) dx$	$(u - 1) \log u$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x) + q(x)} + q(x) \log \frac{2q(x)}{p(x) + q(x)} dx$	$-(u + 1) \log \frac{1+u}{2} + u \log u$
Jensen-Shannon-weighted	$\int p(x) \pi \log \frac{p(x)}{\pi p(x) + (1-\pi)q(x)} + (1-\pi)q(x) \log \frac{q(x)}{\pi p(x) + (1-\pi)q(x)} dx$	$\pi u \log u - (1 - \pi + \pi u) \log(1 - \pi + \pi u)$

f-divergence

$$f^*(t) = \sup_{u \in \text{dom}_f} \{ut - f(u)\}$$

Name	$D_f(P\ Q)$	Generator $f(u)$	$T^*(x)$
Kullback-Leibler	$\int p(x) \log \frac{p(x)}{q(x)} \, dx$	$u \log u$	$1 + \log \frac{p(x)}{q(x)}$
Reverse KL	$\int q(x) \log \frac{q(x)}{p(x)} \, dx$	$-\log u$	$-\frac{q(x)}{p(x)}$
Pearson χ^2	$\int \frac{(q(x)-p(x))^2}{p(x)} \, dx$	$(u-1)^2$	$2\left(\frac{p(x)}{q(x)} - 1\right)$
Squared Hellinger	$\int \left(\sqrt{p(x)} - \sqrt{q(x)}\right)^2 \, dx$	$(\sqrt{u}-1)^2$	$\left(\sqrt{\frac{p(x)}{q(x)}} - 1\right) \cdot \sqrt{\frac{q(x)}{p(x)}}$
Jensen-Shannon	$\frac{1}{2} \int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \, dx$	$-(u+1) \log \frac{1+u}{2} + u \log u$	$\log \frac{2p(x)}{p(x)+q(x)}$
GAN	$\int p(x) \log \frac{2p(x)}{p(x)+q(x)} + q(x) \log \frac{2q(x)}{p(x)+q(x)} \, dx - \log(4)$	$u \log u - (u+1) \log(u+1)$	$\log \frac{p(x)}{p(x)+q(x)}$

f-divergence ELBO

$$\begin{aligned} D_f(P\|Q) &= \int_x q(x) \sup_{t \in \text{dom}_{f^*}} \left\{ t \frac{p(x)}{q(x)} - f^*(t) \right\} dx \\ &\geq \sup_{T \in \mathcal{T}} \left(\int_x p(x) T(x) dx - \int_x q(x) f^*(T(x)) dx \right) \\ &= \sup_{T \in \mathcal{T}} (\mathbb{E}_{x \sim P}[T(x)] - \mathbb{E}_{x \sim Q}[f^*(T(x))]) , \end{aligned}$$

Metrics



Fréchet Inception Distance (FID)

Responsible for FIDelity

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}})$$

Where:

- ▶ μ_r, μ_g : Means of the real and generated image feature distributions.
- ▶ Σ_r, Σ_g : Covariance matrices of the real and generated image feature distributions.
- ▶ Features are extracted using a pre-trained Inception model.

Properties:

- ▶ Lower FID values indicate closer similarity between the distributions.
- ▶ Robust to image transformations like scaling and translations.

Inception Score (IS)

Responsible for devirsity

Mathematical Formulation:

$$\text{IS} = \exp \left(\mathbb{E}_x \left[\text{KL}(p(y|x) \| p(y)) \right] \right)$$

Where:

- ▶ $p(y|x)$: Conditional probability distribution of labels predicted by the Inception model for a generated image x .
- ▶ $p(y)$: Marginal distribution over labels, computed as $p(y) = \int p(y|x)p(x)dx$.
- ▶ $\text{KL}(\cdot \| \cdot)$: Kullback-Leibler divergence.

Interpretation: Higher IS values indicate higher diversity.

Learned Perceptual Image Patch Similarity (LPIPS)

$$\text{LPIPS}(x_1, x_2) = \sum_l w_l \cdot \frac{1}{H_l W_l} \sum_{h,w} \|\phi_l(x_1)_{h,w} - \phi_l(x_2)_{h,w}\|_2^2$$

Where:

- ▶ x_1, x_2 : The two input images being compared.
- ▶ ϕ_l : Features extracted at layer l of a pre-trained network.
- ▶ H_l, W_l : Height and width of the feature map at layer l .
- ▶ w_l : Learned weights for each layer.

Key Features:

- ▶ Correlates with human perception better than traditional metrics like PSNR or SSIM.
- ▶ Designed for tasks where perceptual similarity is more important than pixel-wise similarity.